

$$\vec{a} = \vec{e}_r(\ddot{r} - r\dot{\theta}^2) + \vec{e}_\theta(2\dot{r}\dot{\theta} + \ddot{\theta}r)$$

$$\text{or } \vec{r} = 3\cos 4t \vec{i} + 3\sin 4t \vec{j}$$

$$\vec{V} = -12 \sin 4t \vec{i} + 12 \cos 4t \vec{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -4.8014t \hat{i} + 4.8014t \hat{j}$$

$$\vec{n} = -\sin 4t \vec{i} + \cos 4t \vec{j}$$

$$\therefore \vec{Q}_n = \vec{Q} \cdot \vec{n} = 0$$

$$\vec{a}_t = \vec{a} \cdot \vec{e}_t = -48 \text{ m/s}^2$$

2.

$$\therefore F = ma$$

$$\eta_0 = \int_0^t v \cdot dt$$

$$\therefore kx = m \frac{dx}{dt^2}, \quad \omega = \sqrt{\frac{k}{m}}$$

$$\therefore x = C_1 e^{wt} + C_2 e^{-wt}$$

~~$\phi(t_0) = 0$~~

$$\therefore Q_2 = -Q_1$$

$$\therefore \eta = C(e^{\omega t} - e^{-\omega t})$$

$$v = \frac{dx}{dt} = C\omega (e^{\omega t} + e^{-\omega t})$$

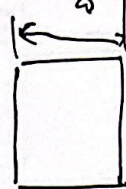
$$V|_{f=0} = 0$$

$$2CW \approx 0$$

10 20 30 40 50 60 70 80 90 100

$$\int_{-\infty}^{\infty} \kappa \, d\kappa = \frac{1}{2} m v^2$$

$$I = mu = \sqrt{kmbo^3}$$



$$L = l_0 \sqrt{1 - \beta^2} \quad , \quad \beta = \left(\frac{v}{c} \right)$$

$$L = a$$

$$\therefore U = C \cdot \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

(4)

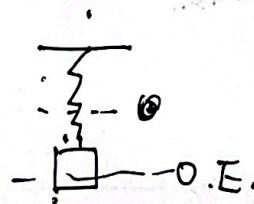
4. 相对论
 $E = mc^2$

$$= \frac{m_0 c^2}{\sqrt{1 - \left(\frac{u}{c}\right)^2}}$$

② 1/4

$$E_2 = E_1 = E_0 = \frac{1}{2} m_0 v^2$$

$$n = \frac{E_2}{E_1} = \frac{\frac{1}{2} m_0 v^2}{\frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}}} = \frac{v^2}{2c^2} \sqrt{1 - \frac{v^2}{c^2}}$$



最大位移为 A .

~~$E = \frac{1}{2} K A^2$~~ $E = \frac{1}{2} K A^2$

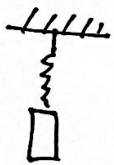
$\therefore$ 当下降时 $\alpha = \frac{1}{2} \text{ GOH}$

~~$$0 = \cancel{mg \frac{A}{2}} + \cancel{\frac{1}{2} k \left(\frac{A}{2} \right)^2} + \frac{1}{2} m v^2$$~~

$\therefore E = \frac{1}{\sqrt{1 - \beta^2}}$

$$E = \frac{1}{2} k \left(\frac{\Delta}{2} \right)^2 + E_k \Rightarrow E_k = \frac{3}{4} E$$





处于平衡位置时

$$k\Delta L = mg$$

而又有:

$$mg - kx = m \frac{dx}{dt}$$

$$-K(x - \Delta L) = m \frac{dx}{dt}$$

$$W = \int \frac{K}{m}$$

$$= \sqrt{\frac{g}{\Delta L}}$$

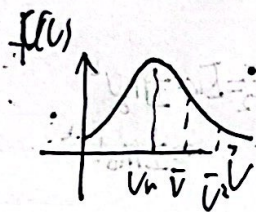
$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{\Delta L}{g}}$$

6.

$$V_1 = \sqrt{\frac{2RT}{M}}$$

$$V_2 = \sqrt{\frac{2RT}{M}}$$

$$V_2 = \sqrt{\frac{1.7RT}{M}}$$



当列解为0时，即总能量不变。

但由于 $T' = 2T_0$

$$\therefore V_2' = \sqrt{2} \cdot V_0$$

即原速率的根号2倍

7.



①. ~~求磁场的磁感强度~~

$$\oint \vec{E} \cdot d\vec{s} =$$

$$dV = \frac{1}{2} \cdot d\epsilon$$

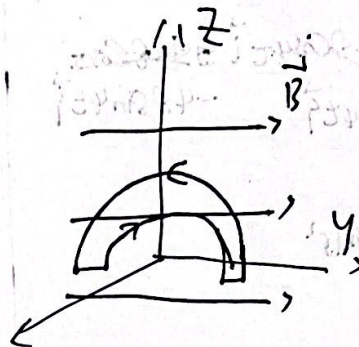
$$= \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} \cdot 2\pi r \cdot dr \cdot \sigma$$

$$= \frac{\sigma}{2\epsilon_0} \cdot dr$$

$$\therefore V = \int_0^R \frac{\sigma}{2\epsilon_0 r} \cdot dr$$

$$= \frac{\sigma}{2\epsilon_0} \ln r \Big|_0^R$$

8.



$$\vec{m} = I \cdot \vec{s}$$

$$= I \cdot \vec{e}_x \cdot (\pi R_2^2 - \pi R_1^2)$$

$$= \frac{\pi I}{2} (R_2^2 - R_1^2) \vec{e}_x$$

$$\vec{M} = \vec{m} \times \vec{B}$$

$$= \frac{\pi I B}{2} (R_2^2 - R_1^2) \cdot \vec{e}_z$$

9.

$$\oint \vec{E} \cdot d\vec{r} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$E = \frac{r}{2} \frac{\partial B}{\partial t}$$

$$= 0.0075 r \cdot \text{方向垂直纸面向里}$$

10.

$$(n_e - n_o) d = \frac{\lambda}{4}$$

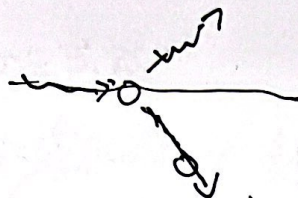
$$d = \frac{\lambda}{4(n_e - n_o)}$$

$$\lambda = \frac{c}{\nu}$$



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11.



$$E_{\lambda} = E_0 + E_{\text{kin}} = 0.5 \text{ MeV}$$

$$E_{\text{kin}} = E_{\lambda} - E_0 = 0.4 \text{ MeV}$$

$$\therefore \frac{E_{\lambda}}{E_{\text{kin}}} = \frac{\frac{hc}{\lambda_{\lambda}}}{\frac{hc}{\lambda_{\text{kin}}}} = \frac{\lambda_{\text{kin}}}{\lambda_{\lambda}} = \frac{5}{4}$$

12.

$$r = n^2 a_0$$

$$\therefore \frac{mv^2}{r} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{e^2}{4\pi\epsilon_0 m r}}$$

$$\therefore p = mv = \sqrt{\frac{e^2 m}{4\pi\epsilon_0 r}}$$

$$\lambda = p = \frac{h}{p} = \frac{2\pi\hbar}{p}$$

$$\therefore \lambda = \frac{2\pi\hbar}{p} = 2\pi\hbar \sqrt{\frac{r}{a_0}}$$

13.

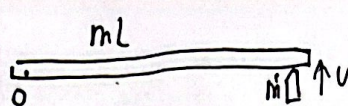


$$m_B g r_B - m_A g r_A = J \alpha$$

$$r_B \alpha = a_A$$

$$m_B = \frac{J \alpha + m_A g r_A}{g r_B r_A}$$

14.



$$J = J_0 + J'$$

$$J_0 = \frac{1}{3} m L^2$$

$$J' = m' L^2$$

$$\therefore J = \frac{1}{3} m L^2 + m' L^2$$

(2) 弹性碰撞中杆的角动量守恒。

故有

$$m' v \cdot L = J \cdot \omega$$

$$\omega = \frac{m' v \cdot L}{J}$$

(3)

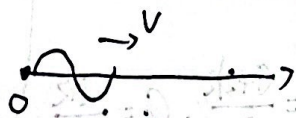
$$W_p = \int_0^{\theta} M_r \cdot d\theta$$

$$= \theta M_r$$

$$W_p = \frac{1}{2} J \cdot \omega^2 = \frac{(m' v L)^2}{2J}$$

$$\therefore \theta = \frac{m'^2 v^2 L^2}{2J m r}$$

15.



$$\omega = \frac{2\pi}{T}$$

$$\omega = \frac{2\pi}{T} = \pi$$

设质点振动表达式为

$$y_0 = A \cos(\pi t + \varphi)$$

$$\text{由图中可知 } A = 10 \text{ cm } y_0|_{t=\frac{1}{2}} = -5 \text{ cm}$$

$$V|_{t=\frac{1}{2}} < 0$$

$$\therefore \text{角初相 } \varphi = \frac{2\pi}{3} \quad y_0 = 10 \cos(\pi t + \frac{2\pi}{3})$$



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(1). 设波的表达式

$$y = 10 \cos(\pi t - (kx + \frac{2\pi}{3}))$$

$$y = 0.1 \cos(\pi t - (kx + \frac{2\pi}{3})) \quad (S2)$$

$$\therefore BC = 2m, \lambda = 4m$$

$$\therefore k = \frac{2\pi}{\lambda} = \frac{\pi}{2}$$

$$\therefore y = 0.1 \cos(\pi t - \frac{\pi}{2}x + \frac{2\pi}{3}) \quad (S2)$$

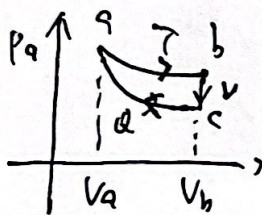
(3). $y = 0$

$$0.1 \cos(\frac{\pi}{3} - \frac{\pi}{2}x + \frac{2\pi}{3}) = 0$$

$$\text{解得 } x = \frac{1}{3} + 2k, \quad k \in \mathbb{Z}$$

$$\therefore x_B = \frac{7}{3} (m)$$

4.



(1) $a \rightarrow b$ 等温过程 $\Delta E = 0$

$$Q = W$$

$$= p_a v_a \ln \frac{v_b}{v_a}$$

$$= \ln 3 \cdot p_a v_a$$

$b \rightarrow c$ 绝热过程 $W = 0$

$$Q_2 = \Delta E_2$$

$$= \frac{m}{M} C_V \Delta T$$

$$= \frac{5}{2} (3^{-\frac{2}{5}} - 1) p_a v_a$$

$$C_p = \frac{5+2R}{2}, \quad C_v = \frac{3R}{2}$$

$$\mu = \frac{C_p}{C_v} = \frac{7}{5}$$

$$T_c v_c^{\mu-1} = T_a v_a^{\mu-1}$$

$$c \rightarrow a, Q_3 = 0$$

$$\therefore Q = Q_1 + Q_2 + Q_3$$

$$= \left[\frac{5}{2} (3^{-\frac{2}{5}} - 1) + \ln 3 \right] p_a v_a$$

(2).

$$\eta = \frac{Q_{12} - Q_{21}}{Q_{12}}$$

$$= \frac{Q_1 - Q_2}{Q_1}$$

$$= \frac{\ln 3 - \frac{5}{2} (3^{-\frac{2}{5}} - 1)}{\ln 3}$$

(3).

$$\Delta S = \int_a^b \frac{dQ}{T}$$

$$= \Delta S = \int_{T_b}^{T_c} \frac{m}{M} C_V \frac{dT}{T}$$

$$= \frac{m}{M} C_V \ln \frac{T_c}{T_b}$$

$$= \frac{m}{M} C_V \ln \frac{T_c}{T_b}$$

$$= -\frac{5}{2} \frac{m}{M} C_V \ln 3$$

5.

$$(1). C = \frac{C_1 C_2}{C_1 + C_2}$$

$$C_1 = \frac{4\pi \epsilon_0 R_1 R_2}{R_2 - R_1}, \quad C_2 = \frac{4\pi \epsilon_0 R_1 R_2}{R_3 - R_2}$$

$$C = \frac{4\pi \epsilon_0 R_1 R_2^2 R_3}{R_1 R_2 (R_3 - R_2) + R_2 R_3 (R_2 - R_1)}$$

$$Q = CU$$

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon} \quad (r_p > R_2)$$

$$E = \frac{Q}{4\pi r^2 \epsilon}$$

$$= \frac{U}{r^2} \cdot \frac{Q R_1 R_2^2 R_3}{R_1 R_2 (R_3 - R_2) + R_2 R_3 (R_2 - R_1)}$$

$$\text{取 } r = R_2 \text{ 代入得 } E = 476.2 \text{ V/m}$$



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(2)

$$\sigma' = \vec{p} \cdot \vec{e}_n$$

$$\begin{aligned} \sigma' &= \oint \vec{p} \cdot d\vec{s} \quad \sigma' = \oint \vec{p} \cdot d\vec{s} \\ &= \oint \vec{D} \cdot \epsilon_0 \vec{E} \cdot d\vec{s} \\ &= \frac{\epsilon - \epsilon_0}{\epsilon} \cdot \oint \vec{D} \cdot d\vec{s} \\ &= \frac{\epsilon - \epsilon_0}{\epsilon} \oint \vec{D} \cdot d\vec{s} \end{aligned}$$

$$\therefore \text{在 } R_1 < r < R_2 \text{ 处 } \vec{e}_s = \vec{e}_r, \vec{e}_n = \vec{e}_r$$

$$\sigma'_1 = -\frac{2}{3}Q = -$$

$$\sigma'_2 = \frac{2}{3}Q = \dots$$

(3)

(1) 在 $R_1 < r < R_2$ 处

$$I' = \frac{\pi(r^2 - R_1^2)}{\pi(R_2^2 - R_1^2)} I$$

$$= \frac{r^2 - R_1^2}{R_2^2 - R_1^2} I$$

$$\therefore \oint \vec{B} \cdot d\vec{l} = \mu_0 I'$$

$$\therefore B = \frac{\mu_0 I'}{2\pi r}$$

$$= \frac{\mu_0 I (r^2 - R_1^2)}{2\pi r (R_2^2 - R_1^2)}$$

(2) 在 $R_2 < r < R_3$ 处

$$\oint \vec{H} \cdot d\vec{l} = I$$

$$\therefore H = \frac{I}{2\pi r}$$

$$B = \frac{\mu_0 \mu_r I}{2\pi r}$$

(3)

$$p \cdot W_e = \frac{1}{2} \vec{H} \cdot \vec{B}$$

$$= \frac{1}{2} \mu H^2$$

$$= \frac{1}{2} \mu \cdot \left(\frac{I}{2\pi r} \right)^2$$

$$= \frac{\mu I^2}{8\pi^2 r^2}$$

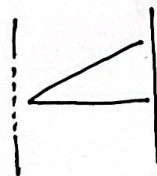
(4)

$$E_e = \int W_e dr$$

$$= \int_{R_2}^{R_3} \frac{\mu I^2}{8\pi^2 r^2} \cdot L \cdot 2\pi r \cdot dr$$

$$= \frac{\mu_0 I^2 L}{4\pi} \ln \frac{R_3}{R_2}$$

7.



(1)

$$d \cdot \sin \theta = k\lambda$$

$$\therefore d = \frac{k\lambda}{\sin \theta}$$

$$\text{已知 } d = 2.8 \times 10^{-6} \text{ m}$$

$$(2) \quad k\lambda =$$

(3) 由



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